Contribution Summary

Unit code: EAT40006
Unit name: Engineering Technology Project B (ENG/CS)-H1
Submission date: 14/10/2026
Contribution for: Sprint 3

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Acknowledgment of Country

I acknowledge the Traditional Owners of the land on which I live, study, and completed this work. I pay my respects to their Elders past and present and acknowledge their continuing connection to Country.

1. CONTRIBUTION EVIDENCE AND IMPACT

1.1 Development and extension of the humanoid control system

Situation

- The project focuses on developing a whole-body Model Predictive Control system for an industrial humanoid robot simulated using the Unitree H1 model in MuJoCo. The initial controller was primarily focused on maintaining a stable standing posture and recovering from external disturbances. The project objective required extending this work toward more realistic humanoid behaviours, particularly walking and stepping.

Task

- My contribution was to extend the existing standing controller toward a walking-capable architecture while maintaining the existing control pipeline of whole-body MPC, joint-space PD control with gravity compensation, and the CBF-based safety filter.

Action

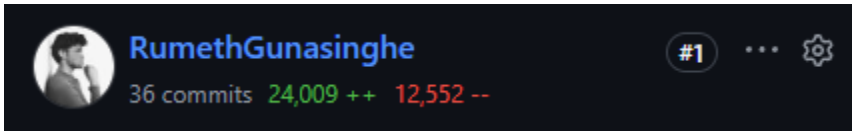
- I investigated the existing controller architecture and implemented and integrated multiple walking components, including:
- A walking state machine for coordinating standing, lateral weight shifting, swing, touchdown, and recovery.
- Lateral weight shifting to move the robot's centre of mass toward the supporting foot before initiating a step.
- Footstep planning for alternating left and right footsteps.
- Swing-foot trajectory generation.
- Analytical leg inverse kinematics for generating joint targets for the swing leg.
- Integration of these components with the existing whole-body MPC, PD controller, gravity compensation, and CBF safety filter.
- Capture-point based checks to prevent the swing phase from beginning when the robot's lateral state was unsuitable.
- Logging and visualisation of walking-related controller behaviour.
- I also investigated the interaction between the walking extension and the existing safety controller rather than treating walking as an independent controller.

Result

- The walking extension successfully progressed beyond the initial standing controller. The robot was able to perform the lateral weight shift and generate a swing trajectory, move the swing foot toward the planned landing location, and reach confirmed foot contact.
- The development also exposed a significant limitation: reaching touchdown did not automatically result in stable double support. This provided a concrete failure case for further research rather than simply assuming that successful foot contact represented successful walking.

Evidence:

Git contribution evidence




```
PS C:\Users\rumet\Desktop\MPC new> git log --format="%h %an %s"
7747e26 RumethGunasinghe added project documents in pdf formats
904a442 RumethGunasinghe Merge branch 'main' of https://github.com/RumethGunasinghe/MPC
fb71b6d RumethGunasinghe foldered all the documents relavent to the project as accordingly
3ca6859 RumethGunasinghe Merge reconstructed project with previous development history
dba3bfb RumethGunasinghe dded another reinforced learning layer to balance the touchdown still need to tune it. tested every unit and documented the outcome, Final sprint completed
1859257 RumethGunasinghe updated readme
7622b9c RumethGunasinghe updated README file
20e7790 RumethGunasinghe try 2
2bd1312 RumethGunasinghe fixing the left foot placement try 2
d5333ea RumethGunasinghe tries to take the left foot off the floor and fails after putting placing back
56776fb RumethGunasinghe com shifting to right for the first step added
1f11dd7 RumethGunasinghe added single step plan and foot step tragectory
2915be6 RumethGunasinghe implemented com shift for walking. (no step tragectory yet)
063abed RumethGunasinghe added visual demonstrations of the model balancing itself after applying a ecternal force
f0cfcbb RumethGunasinghe Project reconstructed with necassary components
196ff18 RumethGunasinghe removed old directories
bdbdf8f RumethGunasinghe Merge branch 'main' of github.com:RumethGunasinghe/MPC
611d642 RumethGunasinghe whatever changed
c5b0540 thinidumadurapperma Add H1 balance controller
81b0978 RumethGunasinghe trying to balance the hips
e2e4939 RumethGunasinghe best balanced yet. stays straight for few seconds
176f593 RumethGunasinghe trying to balance with Nvidia default pose
c5100ed RumethGunasinghe somehwat balaned knees and ankles
4ac53e5 RumethGunasinghe trying to shift the COM to align with ankles
247601b RumethGunasinghe still falling backwards. moving to MPC to implement controlled alanced recovery
2d8d7bb RumethGunasinghe slowed bakcward falling. needs tuning
c8014e7 RumethGunasinghe trying to add graphs to visualize the actions
9440fb8 RumethGunasinghe false slowly now somewhat balanced
6a80c8f RumethGunasinghe Added Knees
8a31934 RumethGunasinghe addded ankle and hip joints. locked all oothers ankles balanced kinda.
636f982 RumethGunasinghe started working on the humanoid robot.
973225d RumethGunasinghe Added Disturbances
8b3295d RumethGunasinghe OKay level finalised ver
8c72075 RumethGunasinghe Working Class based Visualiser that allows to switch between ciontrollers for a Inverse Pendulum
33a2903 RumethGunasinghe balanced cart
bf27c5d RumethGunasinghe Added real physics animation for the pendulum : not tuned
df62bfe RumethGunasinghe added visuals for PID cart  balancing
c5ee102 RumethGunasinghe MPC with constraints for max angle and force
909cbb2 RumethGunasinghe Basic PID and MPC for inverted Pendulum
15392ad RumethGunasinghe Initial commit
PS C:\Users\rumet\Desktop\MPC new>
```

Figure 1 : Git commit history demonstrating individual implementation contributions.

Walking architecture/code evidence

```python
class WalkState(Enum):
    """Discrete phase of the gait."""

    STAND = "STAND"
    SHIFT_LEFT = "SHIFT_LEFT"
    RIGHT_SWING = "RIGHT_SWING"
    TOUCHDOWN_PREP = "TOUCHDOWN_PREP"
    TOUCHDOWN = "TOUCHDOWN"
    DOUBLE_SUPPORT_RECOVERY = "DOUBLE_SUPPORT_RECOVERY"
    DOUBLE_SUPPORT = "DOUBLE_SUPPORT"
    SHIFT_RIGHT = "SHIFT_RIGHT"
    LEFT_SWING = "LEFT_SWING"
```

```python
class LateralWeightShift:
    """Move the centre of mass from double support onto one measured sole.

    ``begin`` is called when a weight shift starts. ``advance`` then rate-limits
    the blend, and ``apply`` writes the blended leg targets over the MPC's
    joint command. The MPC, the PD and the CBF are unchanged; this only
    replaces the leg position targets for the duration of the shift.
    """

    def __init__(self, model) -> None:
        self.model = model
        self.side: str | None = None
        self.fraction = 0.0
        self.origin_xy = np.zeros(2)
        self.target_xy = np.zeros(2)
        self.sole_low = np.zeros(2)
        self.sole_high = np.zeros(2)
        self.pose: dict[str, float] = {}
        self.pelvis_rpy = np.zeros(3)
        self.solved = False
        self.solve_info: dict[str, float] = {}
        # Why the blend did not advance on the last call. "moving" means it did.
        self.limit = "idle"
        self._on_sole = False
        self._com = np.zeros(3)
        # Leg left on its landed pose. apply() must not pull it back to the
        # pre-swing shift solution.
        self.skip_side = ""
        # Late-swing trim. The walking tick turns this on only for a swing,
        # and advance() uses it only while the capture point is closing on
        # the outer sole edge.
        self.capture_trim = False
        self._trim_anchor: float | None = None
        # Touchdown reference. Held so the 8 mm clamp cannot chase the body
        # off the sole after the swing.
        self.handoff_xy: np.ndarray | None = None
        self.hold_landed = False
        # "shift" latches hip roll. "prep" eases the lateral reference inside
        # the 8 mm band. "recovery" leaves hip roll to the MPC capture offset
        # and slews the reference toward recovery_target.
        self.mode = "shift"
        self.reference_speed = 0.15
        self.recovery_target = np.zeros(2)
        self._held_xy: np.ndarray | None = None
        # Debug hold. Recovery normally adds the lateral capture term on top
        # of the shift blend. While this is set, hip roll stays on the blend.
        self.hold_touchdown_hip_roll = False
        self.hip_roll_capture: dict[str, float] = {}
        self.hip_roll_capture_available: dict[str, float] = {}
```

Experimental results

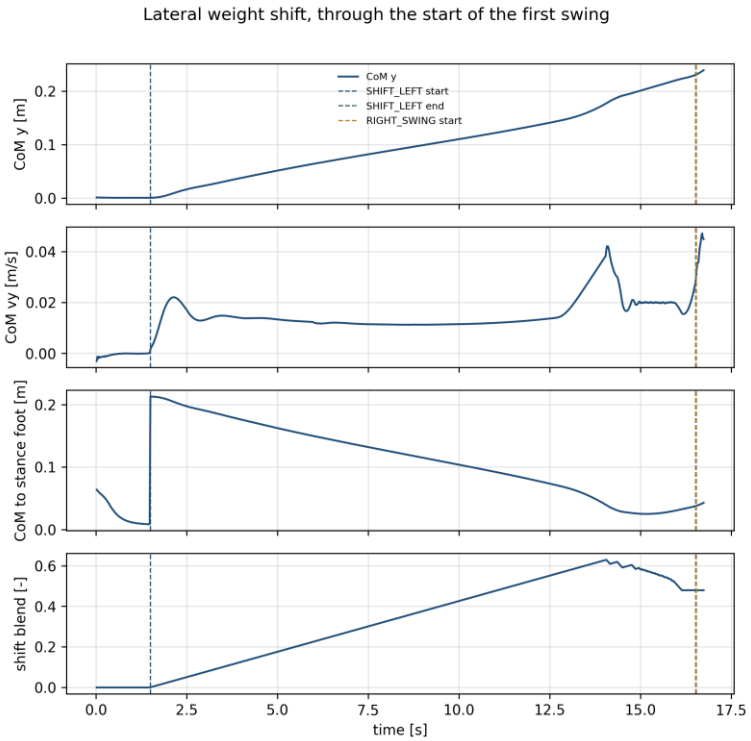


Figure 2 : lateral_weight_shift.png

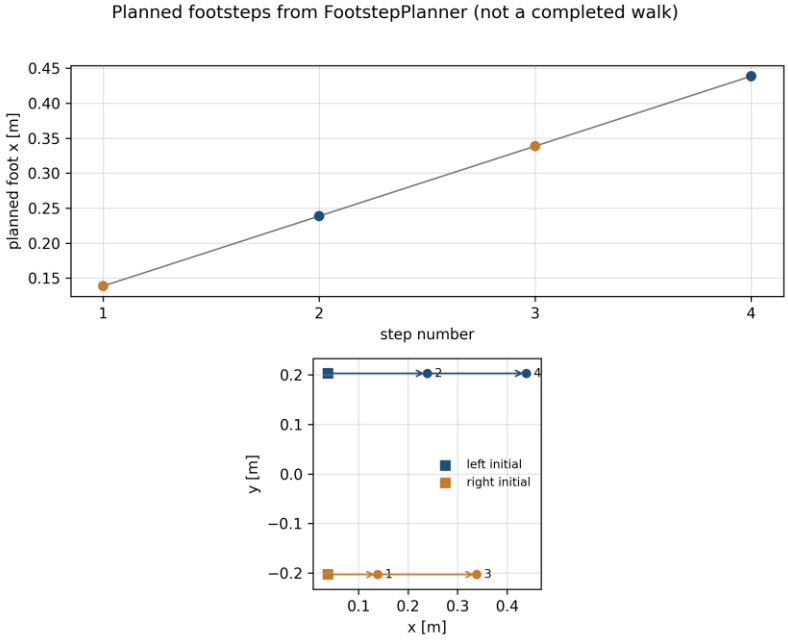


Figure 3 : footstep_plan.png

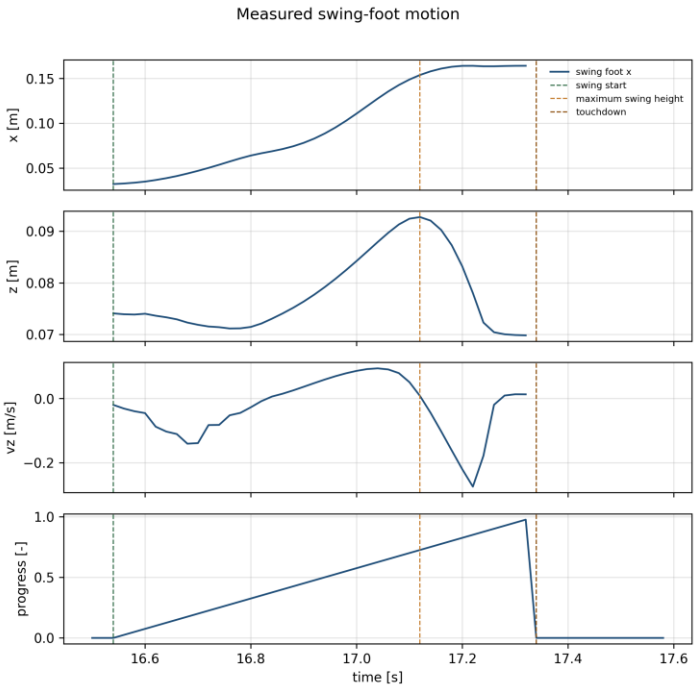


Figure 4 : swing_trajectory.png

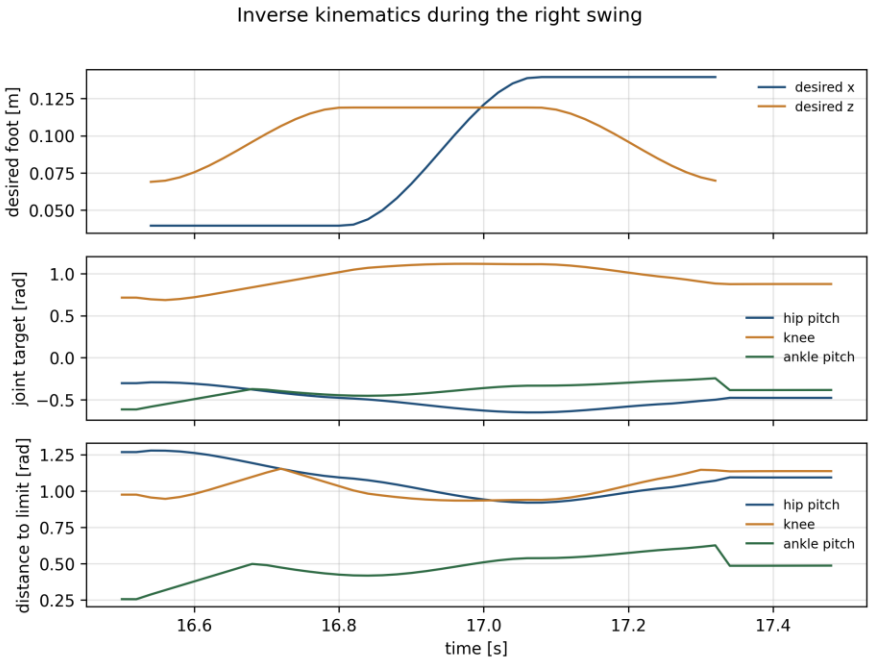


Figure 5 : inverse_kinematics.png

Situation

- After implementing the first walking extension, the robot could complete the swing phase but became unstable during the transition from swing to double support. The robot eventually fell despite the swing foot making contact with the ground.

Task

- I was responsible for investigating the failure rather than simply changing controller parameters without identifying the underlying cause.

Action

- I analysed the simulation logs around the touchdown transition and examined:
    - Joint targets.
    - Joint torques.
    - Foot contact.
    - Foot vertical velocity.
    - Hip-roll commands.
    - Centre-of-mass velocity.
    - Capture-point behaviour.
    - CBF intervention.
    - Support-foot selection.
    - Walking state transitions.
- The investigation identified a discontinuity in the landed ankle's sagittal capture correction. At touchdown, the ankle was receiving an additional approximately −0.15 rad capture correction.
- This produced a significant ankle command change and was associated with a large torque response and an upward velocity change of the landed foot.
- I then modified the controller so that the sagittal capture correction is suppressed for the newly landed foot during TOUCHDOWN and DOUBLE_SUPPORT_RECOVERY, while preserving the calculation and behaviour for the other joints and for standing/push recovery.

Result

- The modification successfully removed the identified touchdown discontinuity.
- Before the fix, the right ankle changed approximately from:
    - −0.244 rad → −0.394 rad at touchdown.
- After the fix, the command remained approximately:
    - −0.244 rad → −0.244 rad
- The previous approximately −49 N·m nominal ankle torque spike was eliminated, and the previous sudden vertical velocity increase was also removed.

- However, the robot still did not achieve stable double support. This was an important result because it demonstrated that the ankle discontinuity was a genuine contributing fault but not the complete cause of the walking failure.

Evidence

Touchdown debugging evidence

Time	Before: ankle command	After: ankle command
17.32 s	≈ -0.244 rad	≈ -0.244 rad
17.34 s	≈ -0.394 rad	≈ -0.244 rad
17.36 s	—	≈ -0.244 rad

Figure 6 : Before/after comparison of the landed-ankle capture correction. The comparison shows that the large ankle-command change immediately after touchdown was removed by suppressing the landed-foot capture correction.

```python
def suppress_landed_ankle_capture(
    offset: np.ndarray,
    joint_names: list[str],
    landed_side: str = "",
    *,
    allow_capture: bool = True,
) -> np.ndarray:
    """Drop the sagittal capture term on the foot that just landed.

    ``offset`` is the clipped capture vector from :meth:`WholeBodyMPC._capture_offset`.
    The stance ankle and both hip rolls are left unchanged. ``allow_capture``
    restores the previous behaviour of adding the term to the landed ankle.
    """
    applied = np.asarray(offset, dtype=float).copy()
    if allow_capture or landed_side not in ("left", "right"):
        return applied
    ankle = f"{landed_side}_ankle"
    for index, name in enumerate(joint_names):
        if name == ankle or str(name).startswith(ankle):
            applied[index] = 0.0
    return applied
```

Figure 7 : landed-ankle suppression.

```python
def _sagittal_capture_offset(offset: np.ndarray, joint_names: list[str]) -> float:
    """Clipped sagittal capture term, read from the first ankle entry."""
    for index, name in enumerate(joint_names):
        if "ankle" in name:
            return float(np.asarray(offset, dtype=float).reshape(-1)[index])
    return 0.0
```

Figure 8 :landed ankle's sagittal capture offset

```
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe -m tests.test_touchdown_recovery
10 passed
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe -m tests.test_touchdown
PASS observation
PASS clip
PASS phase
PASS reward
PASS checkpoint
PASS lagrange_scale
6 passed
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe -m tests.test_lateral_shift
PASS planners
PASS leg_ik
PASS solver
PASS stand
PASS walk_stays_up
PASS target_moves
6 passed
```

Figure 9 : Terminal screenshot showing the test results.

1.2 Investigation of Double-Support Recovery and Load Transfer

Situation

- After the ankle discontinuity was removed, the robot continued to fall during double-support recovery. This indicated that the problem was not simply an impact or ankle-target problem.

Task

- I continued the investigation to determine why the newly landed foot was unable to become a stable load-bearing support.

Action

- I analysed the normal contact forces of both feet and the recovery hip-roll behaviour.
- The analysis showed that the recovery controller initially changed the hip-roll command abruptly when entering DOUBLE_SUPPORT_RECOVERY. The command changed by approximately 0.15 rad in a single control tick because the recovery controller added the lateral capture correction immediately.
- I created a controlled experiment in which the touchdown hip-roll command was held at its pre-touchdown value.

- This experiment was intentionally isolated so that the ankle suppression and other controller components remained unchanged.
- I then compared:
    - Normal recovery.
    - Hip-roll hold recovery.
    - The previous ankle-fix experiment.

Result

- Holding the hip-roll command increased the duration of right-foot contact from approximately:
    - 17.26–17.46 s to :17.26–17.80 s.
- This demonstrated that the abrupt hip-roll transition was contributing to early unloading of the newly landed foot.
- However, the experiment also revealed a more fundamental issue.
- The right foot did **not** progressively take the robot's weight. For example, during the held-hip-roll experiment:

Time	Right force	Left force
17.32 s	135 N	418 N
17.34 s	87 N	437 N
17.36 s	66 N	423 N
17.40 s	33 N	422 N
17.80 s	8 N	511 N

- Therefore, even when the hip-roll transition was removed, the left foot continued carrying almost the entire load while the right foot progressively unloaded.
- The robot ultimately lost right-foot contact and fell.

Impact

- This investigation narrowed the remaining failure from a general "walking instability" problem to a more specific **load-transfer and planted-foot control problem**.
- The evidence indicates that:
- The right foot is geometrically touching the ground but is not successfully becoming a load-bearing support foot.

Evidence

```
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe main_demo.py --headless --steps 1 --duration 24 --seed 0 --log-path data\logs\walking_recovery_hiproll_hold_v1.npz
[2026-10-15 00:09:14,517] INFO    main_demo | headless run. 1 forward steps, centre of mass shifts onto the stance foot before each swing.
[2026-10-15 00:09:29,092] INFO    main_demo | stopped: pelvis height 0.701 m
[2026-10-15 00:09:29,296] INFO    main_demo | closed after 18.68 s, 934 steps. peak intervention 133.81 Nm. log: data\logs\walking_recovery_hiproll_hold_v1.npz
=== SINGLE SUPPORT DIAGNOSTIC ===

Swing: right
Stance: left
Swing duration: 0.58 s (16.54 s to 17.12 s)

First support_x_lower <= 0: none
First support_x_lower < 0.02 m: none
First CBF not optimal: t = 18.00 s (DOUBLE_SUPPORT_RECOVERY)
First backward CoM velocity < -0.05 m/s: t = 17.28 s (TOUCHDOWN_PREP)

Minimum support_x_lower during swing: 0.0617 m
Maximum backward CoM velocity during swing: -0.0079 m/s
Maximum backward CoM acceleration during swing: -0.0546 m/s^2
Minimum capture-point rear margin during swing: 0.0594 m
Maximum |stance ankle correction|: 0.3210 rad
Maximum stance knee movement from swing start: 0.0191 rad
Maximum |swing-leg torque|: 17.7547 N.m
Maximum |swing knee acceleration|: 10.0580 rad/s^2
At swing end capture (+0.0529, +0.2329) m

Landing window: 17.14 s to 18.62 s
Maximum outward CoM velocity after swing: 1.3639 m/s
Minimum outer support margin after swing: -0.4586 m
Shift fraction at landing: 0.4790
Shift fraction 0.4 s later: 0.4790

Touchdown (first tick after the swing): t = 17.14 s
time | state | posture | com xy | v xy | capture xy | ref xy | dref | left hip-roll des | ankle corr | fraction | cbf | iv | xl | outer y | cap rear | cap outer | pitch
 16.94 | RIGHT_SWING      | on   | (+0.056,+0.245) | (-0.006,+0.003) | (+0.055,+0.246) | (+0.064,+0.237) | 0.003 | -0.155 | -0.314 | 0.479 | optimal |    0.0 | +0.063 | +0.050 | +0.061 | +0.049 | -0.123
 17.14 | TOUCHDOWN_PREP   | on   | (+0.055,+0.241) | (-0.008,-0.031) | (+0.053,+0.232) | (+0.063,+0.234) | 0.003 | -0.155 | -0.323 | 0.479 | optimal |   28.0 | +0.062 | +0.054 | +0.059 | +0.064 | -0.132
 17.16 | TOUCHDOWN_PREP   | on   | (+0.055,+0.241) | (-0.009,-0.032) | (+0.052,+0.231) | (+0.063,+0.233) | 0.003 | -0.155 | -0.326 | 0.479 | optimal |   35.6 | +0.061 | +0.055 | +0.059 | +0.064 | -0.133
 17.24 | TOUCHDOWN_PREP   | on   | (+0.054,+0.238) | (-0.012,-0.020) | (+0.051,+0.232) | (+0.062,+0.231) | 0.003 | -0.155 | -0.325 | 0.479 | optimal |   84.0 | +0.061 | +0.057 | +0.057 | +0.063 | -0.132
 17.34 | TOUCHDOWN        | on   | (+0.049,+0.241) | (-0.071,+0.045) | (+0.028,+0.255) | (+0.058,+0.232) | 0.004 | -0.155 | -0.369 | 0.479 | optimal |    0.0 | +0.056 | +0.054 | +0.035 | +0.040 | -0.140
 17.54 | DOUBLE_SUPPORT_RECOVERY | on   | (+0.040,+0.257) | (-0.019,+0.105) | (+0.034,+0.288) | (+0.067,+0.213) | 0.021 | -0.155 | -0.402 | 0.479 | optimal |    0.0 | +0.049 | +0.039 | +0.044 | +0.087 | -0.156
Dense window every 0.04 s:
 16.90 RIGHT_SWING      v (-0.007,+0.014) y +0.245 capOut +0.046 yOut +0.050 hip -0.155/-0.160 ank -0.716/-0.434 knee +1.102 frac 0.479 com_inside
 16.94 RIGHT_SWING      v (-0.006,+0.003) y +0.245 capOut +0.049 yOut +0.050 hip -0.155/-0.160 ank -0.714/-0.407 knee +1.114 frac 0.479 com_inside
 16.98 RIGHT_SWING      v (-0.005,-0.007) y +0.245 capOut +0.053 yOut +0.051 hip -0.155/-0.160 ank -0.712/-0.376 knee +1.117 frac 0.479 com_inside
 17.02 RIGHT_SWING      v (-0.005,-0.017) y +0.245 capOut +0.056 yOut +0.051 hip -0.155/-0.160 ank -0.711/-0.348 knee +1.114 frac 0.479 com_inside
 17.06 RIGHT_SWING      v (-0.006,-0.025) y +0.244 capOut +0.060 yOut +0.052 hip -0.155/-0.160 ank -0.714/-0.333 knee +1.112 frac 0.479 com_inside
 17.10 RIGHT_SWING      v (-0.007,-0.029) y +0.243 capOut +0.062 yOut +0.053 hip -0.155/-0.160 ank -0.719/-0.331 knee +1.107 frac 0.479 com_inside
 17.14 TOUCHDOWN_PREP   v (-0.008,-0.031) y +0.241 capOut +0.064 yOut +0.054 hip -0.155/-0.160 ank -0.723/-0.322 knee +1.078 frac 0.479 touchdown_prep
 17.18 TOUCHDOWN_PREP   v (-0.009,-0.032) y +0.240 capOut +0.065 yOut +0.055 hip -0.155/-0.160 ank -0.727/-0.306 knee +1.035 frac 0.479 touchdown_prep
 17.22 TOUCHDOWN_PREP   v (-0.008,-0.030) y +0.239 capOut +0.065 yOut +0.056 hip -0.155/-0.160 ank -0.727/-0.287 knee +0.989 frac 0.479 touchdown_prep
 17.26 TOUCHDOWN_PREP   v (-0.032,+0.014) y +0.238 capOut +0.053 yOut +0.057 hip -0.155/-0.160 ank -0.739/-0.271 knee +0.950 frac 0.479 touchdown_prep
 17.30 TOUCHDOWN_PREP   v (-0.054,+0.040) y +0.240 capOut +0.044 yOut +0.055 hip -0.155/-0.160 ank -0.756/-0.255 knee +0.905 frac 0.479 touchdown_prep
 17.34 TOUCHDOWN        v (-0.071,+0.045) y +0.241 capOut +0.040 yOut +0.054 hip -0.155/-0.160 ank -0.769/-0.244 knee +0.882 frac 0.479 touchdown_prep
 17.38 TOUCHDOWN        v (-0.059,+0.055) y +0.243 capOut +0.036 yOut +0.052 hip -0.155/-0.160 ank -0.781/-0.244 knee +0.882 frac 0.479 touchdown_prep
 17.42 DOUBLE_SUPPORT_RECOVERY v (-0.050,+0.073) y +0.246 capOut +0.028 yOut +0.049 hip -0.155/-0.160 ank -0.778/-0.236 knee +0.877 frac 0.479 recovery
 17.46 DOUBLE_SUPPORT_RECOVERY v (-0.040,+0.086) y +0.249 capOut +0.020 yOut +0.046 hip -0.155/-0.160 ank -0.788/-0.236 knee +0.878 frac 0.479 recovery
 17.50 DOUBLE_SUPPORT_RECOVERY v (-0.030,+0.096) y +0.253 capOut +0.014 yOut +0.043 hip -0.155/-0.160 ank -0.796/-0.236 knee +0.878 frac 0.479 recovery
[2026-10-15 00:09:30,369] INFO    main_demo | support plot: data\logs\walking_recovery_hiproll_hold_v1_support.png
PS C:\Users\rumet\Desktop\MPC new\project>
```

Figure 10 : The right foot is physically touching the ground, but its normal force rapidly decreases, while the left foot continues carrying most of the load.

```python
def _foot_normal_force(env: IndustrialHumanoidEnv, side: str) -> float:
    """Sum of MuJoCo contact-frame normal forces on ``side``.

    The H1 sole capsules are unnamed children of the ankle body, so a contact
    counts when its geom name starts with the side or its body is that ankle.
    The value is the compressive normal reported by ``mj_contactForce``.
    """
    import mujoco

    body_id = mujoco.mj_name2id(env.model, mujoco.mjtObj.mjOBJ_BODY, f"{side}_ankle_link")
    total = 0.0
    wrench = np.zeros(6)
    for index in range(int(env.data.ncon)):
        contact = env.data.contact[index]
        if float(contact.dist) > 1e-3:
            continue
        geoms = (int(contact.geom1), int(contact.geom2))
        names = [
            mujoco.mj_id2name(env.model, mujoco.mjtObj.mjOBJ_GEOM, geom) or "" for geom in geoms
        ]
        bodies = [int(env.model.geom_bodyid[geom]) for geom in geoms]
        named = [name.startswith(f"{side}_") for name in names]
        owned = [body == body_id for body in bodies] if body_id >= 0 else [False, False]
        own = [name_hit or body_hit for name_hit, body_hit in zip(named, owned)]
        if any(own) and not all(own):
            mujoco.mj_contactForce(env.model, env.data, index, wrench)
            total += float(wrench[0])
    return total
```

Figure 11 : Code normal-force logging

Double-support load-transfer analysis

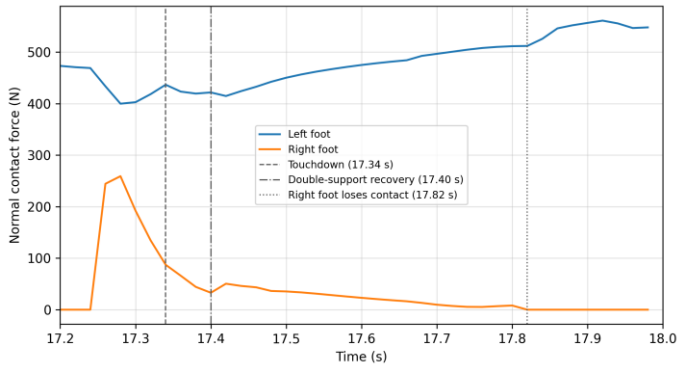


Figure 12 :left and right normal force from 17.2 s to 18.0 s, with lines at touchdown (17.34 s), recovery (17.40 s), and contact loss (17.82 s).

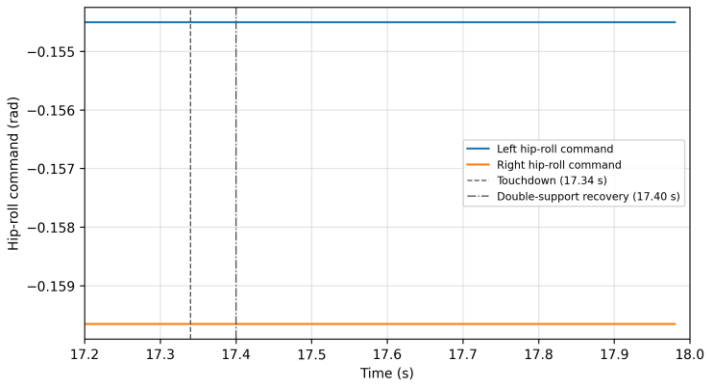


Figure 13 :left and right hip-roll commands over the same window, with lines at 17.34 s and 17.40 s. Both commands stay flat.

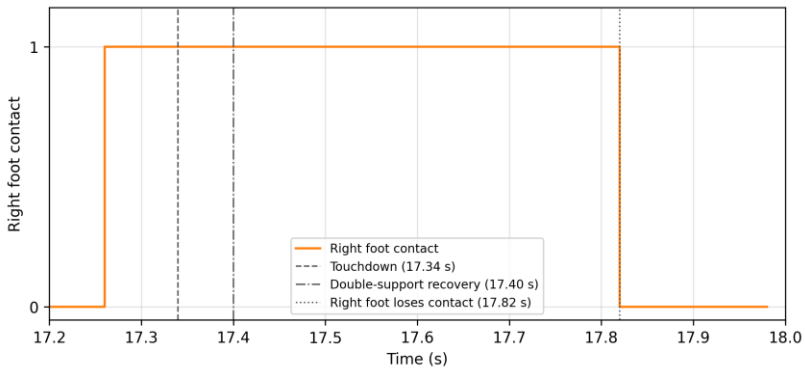


Figure 14 :right-foot contact as 0/1, with lines at 17.34 s, 17.40 s, and 17.82 s.

1.3 Development of Controlled Recovery Experiments

Situation

- Changing several controller parameters simultaneously would make it difficult to determine which modification affected the walking behaviour.

Task

- I needed to establish a controlled debugging methodology where individual hypotheses could be tested independently.

Action

- I performed isolated experiments targeting specific failure mechanisms.
- The experiments included:
- Investigation of the landed ankle capture correction.
- Suppression of the landed ankle sagittal capture correction.
- Investigation of the recovery hip-roll transition.
- Holding the hip-roll command through touchdown/recovery.
- Comparison of foot contact duration and normal force.
- Logging of CBF intervention and recovery behaviour.
- I deliberately kept unrelated controller components unchanged between experiments.

Result

- The experiments progressively eliminated possible causes and produced a clearer understanding of the failure:
    - Walking failure
    - Touchdown instability
    - Landed ankle discontinuity
    - Ankle discontinuity removed
    - Robot still falls
    - Recovery hip-roll step identified
    - Hip-roll hold extends contact
    - Robot still falls
    - Right foot does not receive sufficient load
    - Remaining problem = load transfer / planted-foot support
- This systematic process reduced the risk of introducing unrelated changes that could make the experimental results difficult to interpret.

Evidence

Experiment comparison

Experiment	Main change	Right-foot contact	Stable DS	Conclusion
Baseline	Original walking	~17.26–17.46 s	No	Landing instability
Ankle fix	Suppress landed ankle capture	~17.26–17.46 s	No	Ankle discontinuity removed but not sufficient
Hip-roll hold	Hold pre-touchdown hip roll	~17.26–17.80 s	No	Hip-roll step causes early unloading but isn't root cause

Figure 15 :this shows analysis, experimentation and engineering judgementwith data taken from .npz files

1.4 Testing and Validation

Situation

- The walking extension introduced several interacting components, meaning changes could potentially break previously working functionality.

Task

- I maintained and expanded tests for the walking and recovery components while modifying the controller.

Action

- I ran the relevant automated tests after controller changes and added recovery-specific tests to verify the new behaviour.
- The tests covered:
- Lateral weight shifting.
- Touchdown behaviour.
- Double-support recovery.
- Hip-roll recovery behaviour.
- Preservation of the calculated but suppressed capture term.

Result

- The latest verified test results were:
    - Recovery tests: 10 passed
    - Touchdown tests: 6 passed
    - Lateral shift tests: 6 passed
- The tests confirmed that the ankle suppression and recovery modifications did not break the existing tested components.
- However, I did not treat passing unit/component tests as proof that walking was solved. The full simulation continued to demonstrate failure during double-support recovery.

Evidence

Testing evidence

```
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe -m tests.test_touchdown_recovery
10 passed
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe -m tests.test_touchdown
PASS observation
PASS clip
PASS phase
PASS reward
PASS checkpoint
PASS lagrange_scale
6 passed
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe -m tests.test_lateral_shift
PASS planners
PASS leg_ik
PASS solver
PASS stand
PASS walk_stays_up
PASS target_moves
6 passed
PS C:\Users\rumet\Desktop\MPC new\project>
```

Figure 16 : Automated validation of touchdown recovery, touchdown and lateral weight-shifting components.

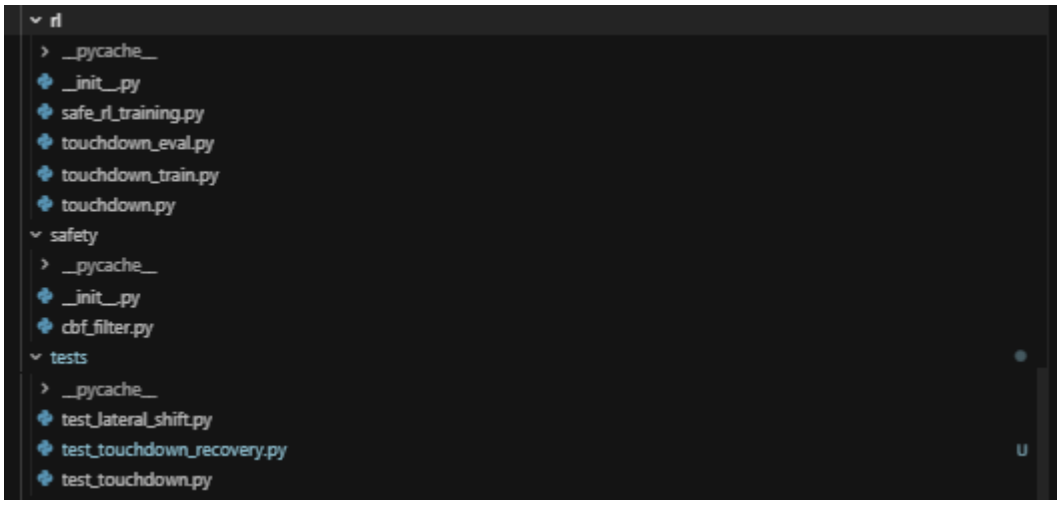


Figure 17 Relevant testing files

1.5 Performance Evaluation of the Existing Controller

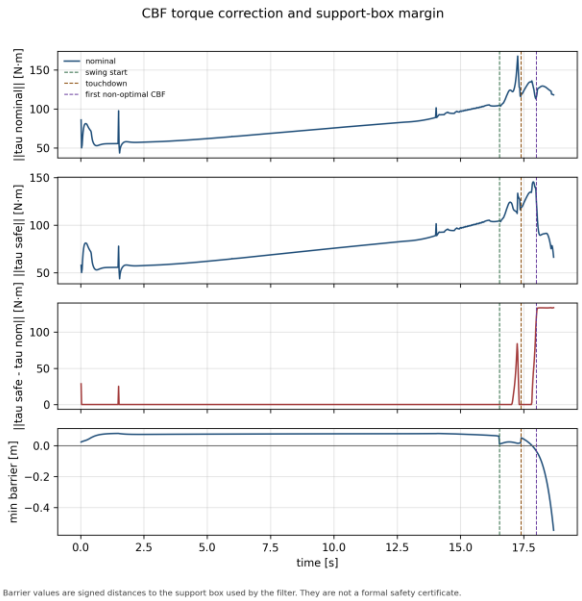
- In addition to walking development, I evaluated the existing standing controller to establish a reliable baseline.
- The controller was tested using an 80 N forward disturbance applied to the torso for 0.2 seconds.
- The controller maintained an upright posture and recovered from the disturbance.
- Measured results included approximately:
- Minimum pelvis height: **0.927 m**
- Final pelvis height: **0.928 m**
- Peak torso pitch: **0.218 rad / 12.5°**
- Peak CBF intervention: approximately **120 N·m**
- The MPC solve-time analysis also showed that the controller's average solve time was approximately **0.95 ms**, with a 95th percentile of approximately **1.71 ms**, compared with the 20 ms control period.

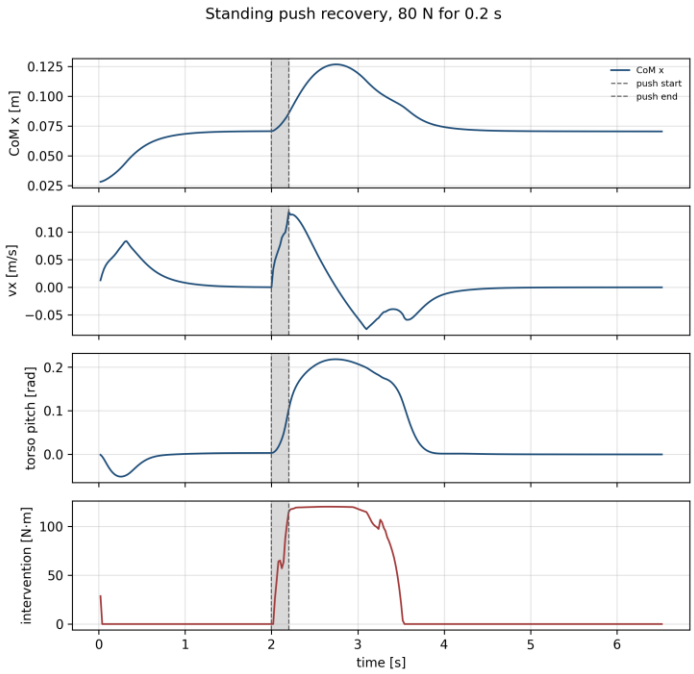
Evidence

Standing controller validation

```
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe main_demo.py --no-walk --headless --duration 6.5 --seed 0 --log-path data\logs\standing_push_latest.npz
[2026-10-15 00:17:52,479] INFO    main_demo | headless run. Push of 80 N at t = 2.0 s.
[2026-10-15 00:17:53,877] INFO    main_demo | t = 2.00 s: applied [80.  0.  0.] N for 0.20 s on torso_link
[2026-10-15 00:17:57,626] INFO    main_demo | closed after 6.52 s, 326 steps. peak intervention 120.27 Nm. log: data\logs\standing_push_latest.npz
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe utils\generate_project_figures.py --log data\logs\standing_push_latest.npz
figures written: 1
  fig_push_recovery.png
PS C:\Users\rumet\Desktop\MPC new\project> ..\.venv\Scripts\python.exe main_demo.py --no-walk
[2026-10-15 00:18:23,425] INFO    main_demo | viewer open — close the window to stop. Push of 80 N at t = 2.0 s.
[2026-10-15 00:18:25,719] INFO    main_demo | t =  1.00 s | STAND support both | COM [0.068 0.000 0.932] m | ref [0.079 0.000] m | stance [0.000 0.000] m | dist 0.000 m | error [-0.011 0.000 -0.018] m | |v| = 0.008 m/s | intervention 0.00 Nm (optimal)
[2026-10-15 00:18:26,871] INFO    main_demo | t =  2.00 s | STAND support both | COM [0.071 0.000 0.931] m | ref [0.079 0.000] m | stance [0.000 0.000] m | dist 0.000 m | error [-0.009 0.000 -0.019] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-10-15 00:18:26,871] INFO    main_demo | t = 2.00 s: applied [80.  0.  0.] N for 0.20 s on torso_link
[2026-10-15 00:18:28,559] INFO    main_demo | t =  3.02 s | STAND support both | COM [0.118 0.000 0.957] m | ref [0.079 0.000] m | stance [0.000 0.000] m | dist 0.000 m | error [0.039 0.000 0.006] m | |v| = 0.060 m/s | intervention 117.30 Nm (relaxed)
[2026-10-15 00:18:29,948] INFO    main_demo | t =  4.02 s | STAND support both | COM [0.074 0.000 0.930] m | ref [0.079 0.000] m | stance [0.000 0.000] m | dist 0.000 m | error [-0.006 0.000 -0.021] m | |v| = 0.012 m/s | intervention 0.00 Nm (optimal)
[2026-10-15 00:18:30,688] INFO    main_demo | closed after 4.50 s, 225 steps. peak intervention 120.27 Nm. log: C:\Users\rumet\Desktop\MPC new\project\data\demo_log.npz
PS C:\Users\rumet\Desktop\MPC new\project>
```

Figure 18 : Terminal output showing the standing experiment





push_recovery_summary.json M

project > data > figures > {} push_recovery_summary.json


```
{
  "log": "data\\logs\\standing_push_latest.npz",
  "push_force_n": [
    80.0,
    0.0,
    0.0
  ],
  "push_window_s": [
    2.0,
    2.2
  ],
  "peak_pitch_rad": 0.2182793412168029,
  "peak_pitch_deg": 12.506485006618801,
  "peak_com_vx_m_s": 0.13639992019760566,
  "peak_com_speed_m_s": 0.17720763406385112,
  "max_com_x_displacement_m": 0.09836195647894061,
  "min_com_x_displacement_m": 0.0,
  "peak_cbf_intervention_nm": 120.26912150704132,
  "final_com_m": [
    0.0704751328038,
    0.00037256543753263853,
    0.9284789871102664
  ],
  "final_torso_pitch_rad": -0.00038143502954205176,
  "final_robot_height_m": 0.9284789871102664
}
```


Figure 19 : generated JSON summary for recovery

1.7 Documentation and Project Artefacts

- I also contributed to documenting the implementation and experimental findings so that the development could be understood and reproduced by the team.
- This included documenting:
    - Controller architecture.
    - Walking state transitions.
    - Experimental configuration.
    - Failure behaviour.
    - Testing results.
    - Simulation commands.
    - Generated figures.
    - Limitations and future work.
- I maintained reproducible simulation commands such as:
    - ..\.venv\Scripts\python.exe main_demo.py --no-walk --headless
- for the standing controller and:
    - ..\.venv\Scripts\python.exe main_demo.py --headless --steps 1 --duration 24 --seed 0
- for the one-step walking experiment.

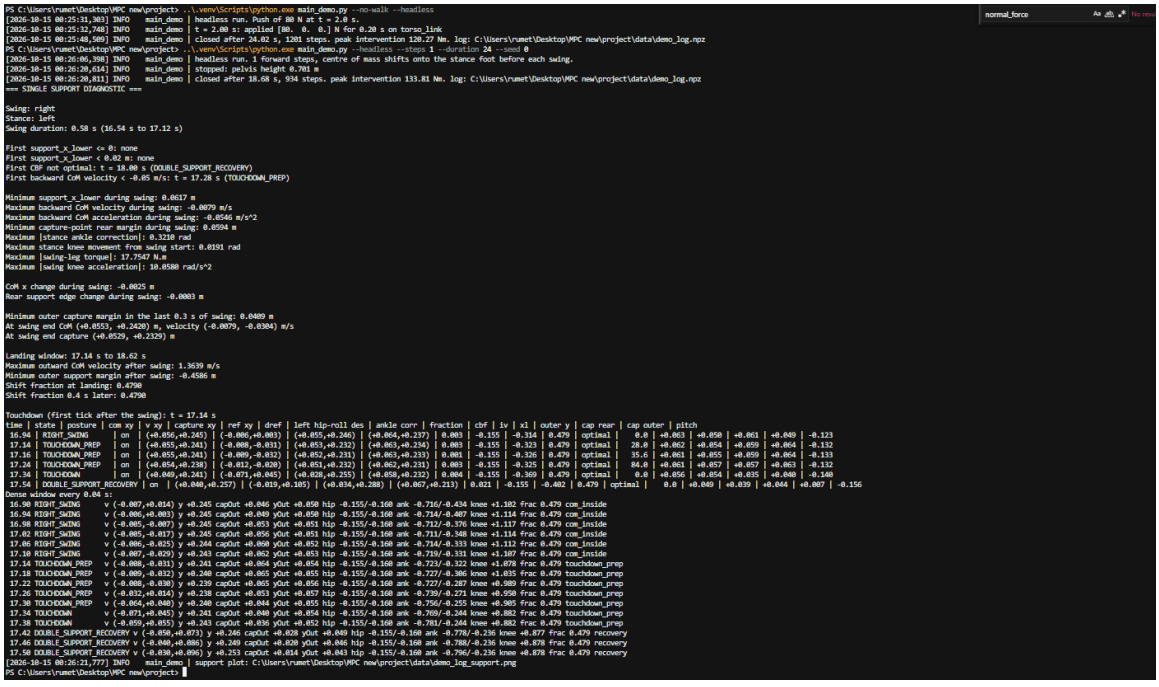


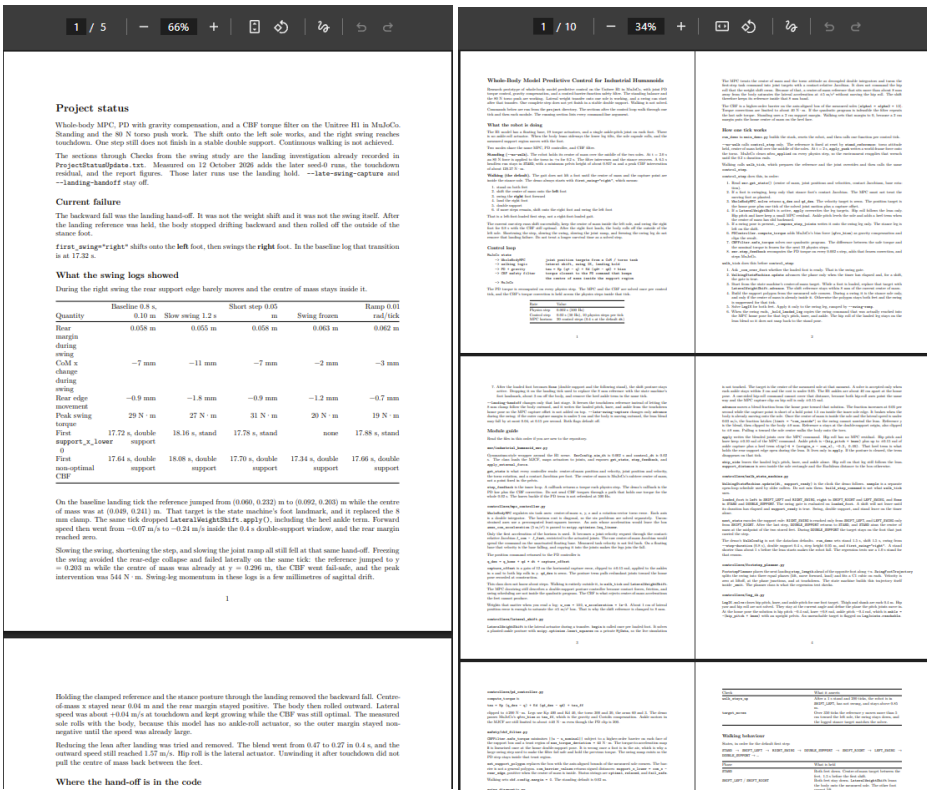
Figure 20 :simulation commands

Evidence

Documentation evidence

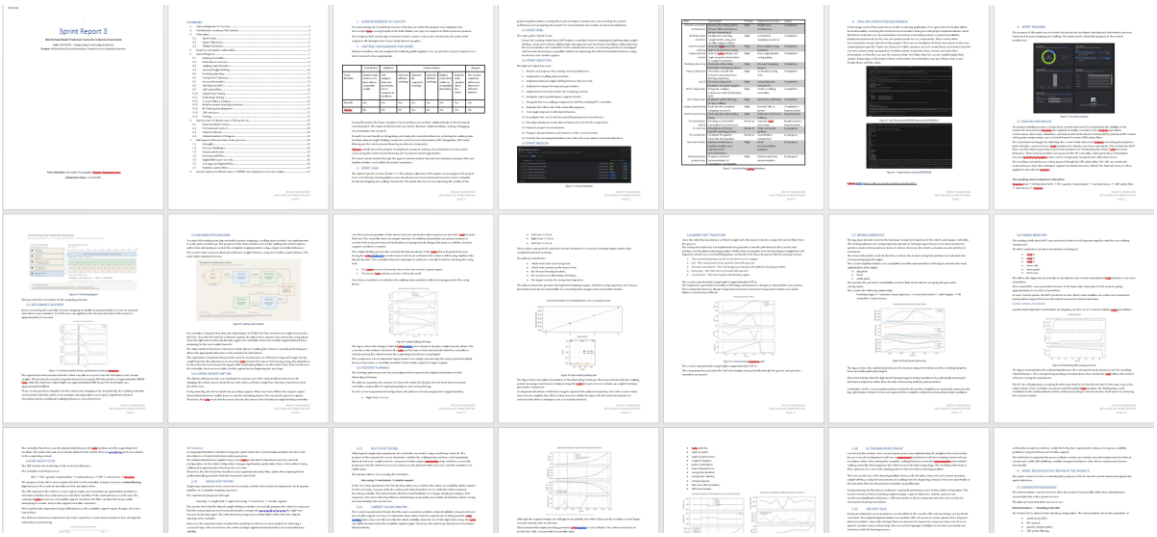


Relevant GitHub README/documentation screenshots.

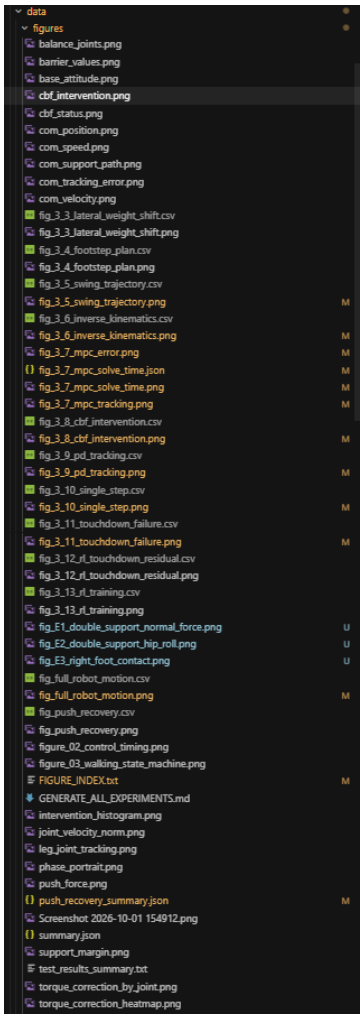


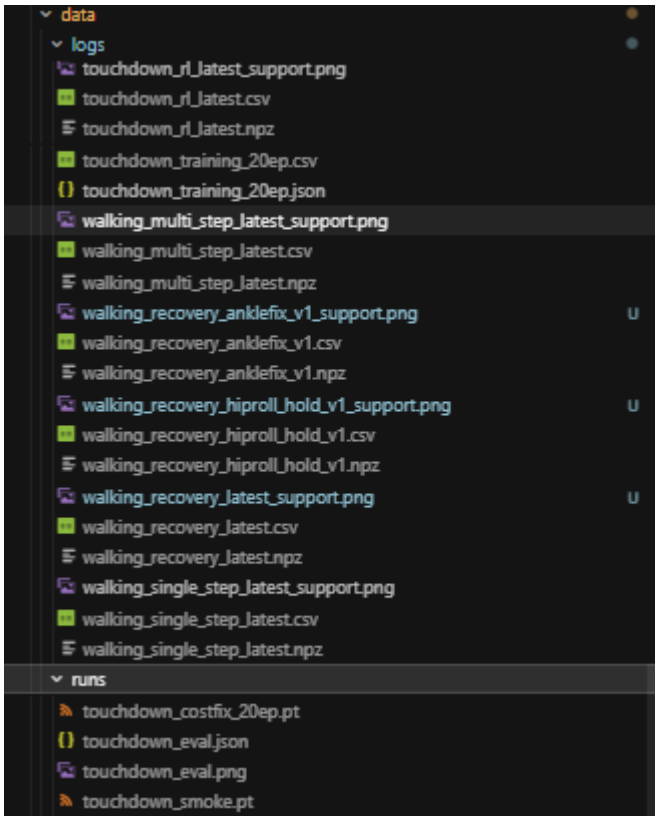
Relevant documentation screenshots done by me.

Final sprint report 3 was totally done by myself



- Experiment-generation documentation and figure generation.





1.6 Summary of Individual Contribution

- My contribution during this sprint/phase focused primarily on **implementation, integration, experimental debugging, validation and documentation of the humanoid walking extension and report creation/submission**.
- The most significant contribution was not only implementing additional walking functionality but also systematically investigating why the resulting controller did not achieve stable double support.
- The investigation progressed from a general walking failure to several specific findings:
    - The landed ankle received an unwanted sagittal capture correction.
    - Suppressing this correction removed the touchdown ankle discontinuity.
    - The robot nevertheless continued to fall.
    - The recovery hip-roll transition caused early unloading of the newly landed foot.
    - Holding the hip roll extended foot contact but did not achieve load transfer.

    - Normal-force measurements showed that the newly landed foot was touching the ground without becoming a significant load-bearing support.
    - The remaining problem was therefore narrowed toward double-support load transfer and planted-foot control.
- This work provided a clear technical foundation for the next development stage.

Total time spent

- **Total time spent on the project during this sprint/phase: 110hours.**

2. TEAMWORK, COMMUNICATION AND PROFESSIONAL PRACTICE=

During this phase, I worked collaboratively on the development and evaluation of the humanoid control system.

I communicated technical progress and experimental findings with the project team and used reproducible simulation results to discuss controller behaviour rather than relying only on subjective observations.

When failures occurred, I investigated the underlying behaviour through simulation logs, controller outputs and test results. This allowed technical discussions to focus on measurable evidence such as joint commands, contact states, normal forces, centre-of-mass velocity and CBF intervention.

I also maintained communication regarding project deliverables, client interaction and project documentation.

Where implementation tasks overlapped with other project work, I separated my contribution by identifying the specific controller components, debugging work and experiments that I was responsible for.

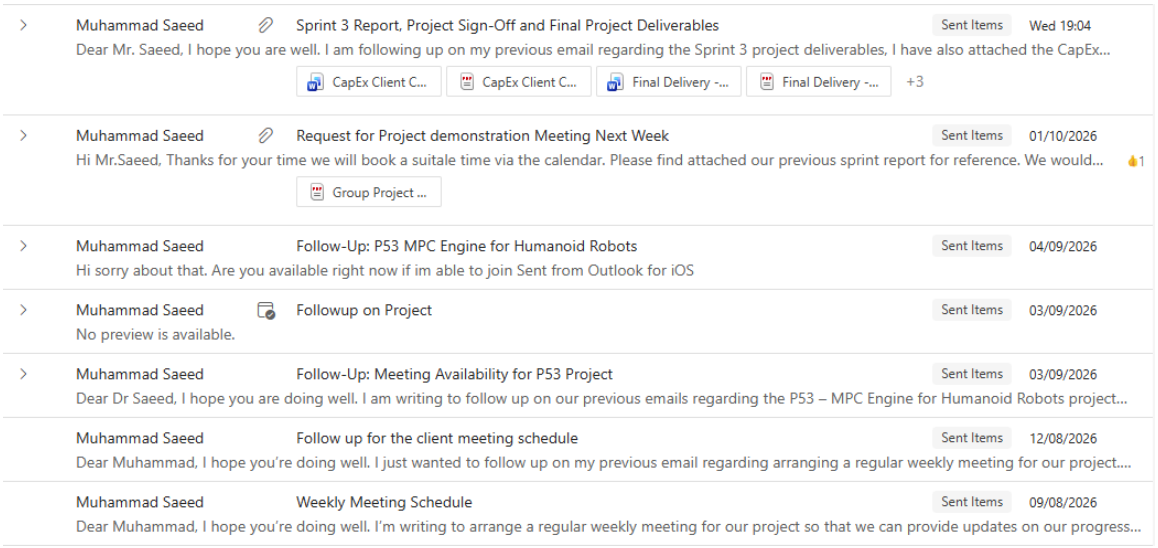
The development process also involved documenting unsuccessful experiments. Rather than presenting only successful outcomes, I recorded cases where a modification improved one metric but did not solve the overall walking problem.

For example, holding the recovery hip-roll command increased right-foot contact duration but did not achieve stable double support. Recording this result prevented the team from incorrectly treating increased contact duration as a successful walking solution.

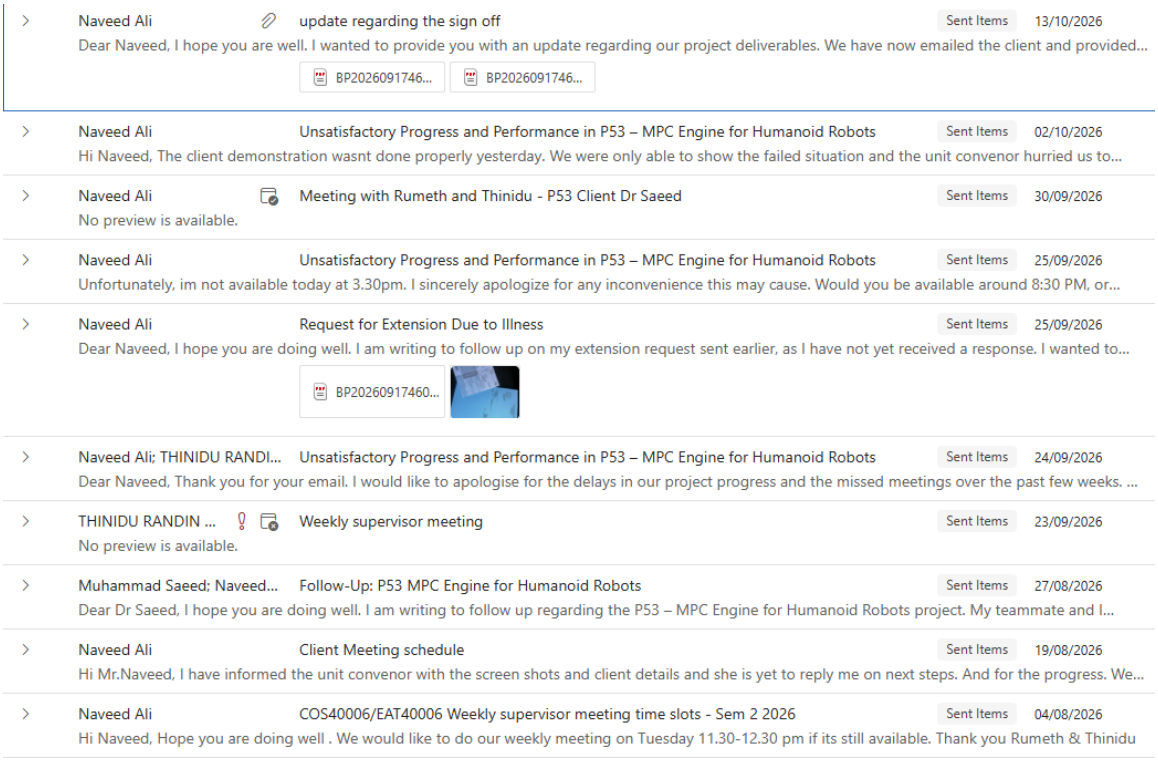
Evidence

Communication and teamwork evidence

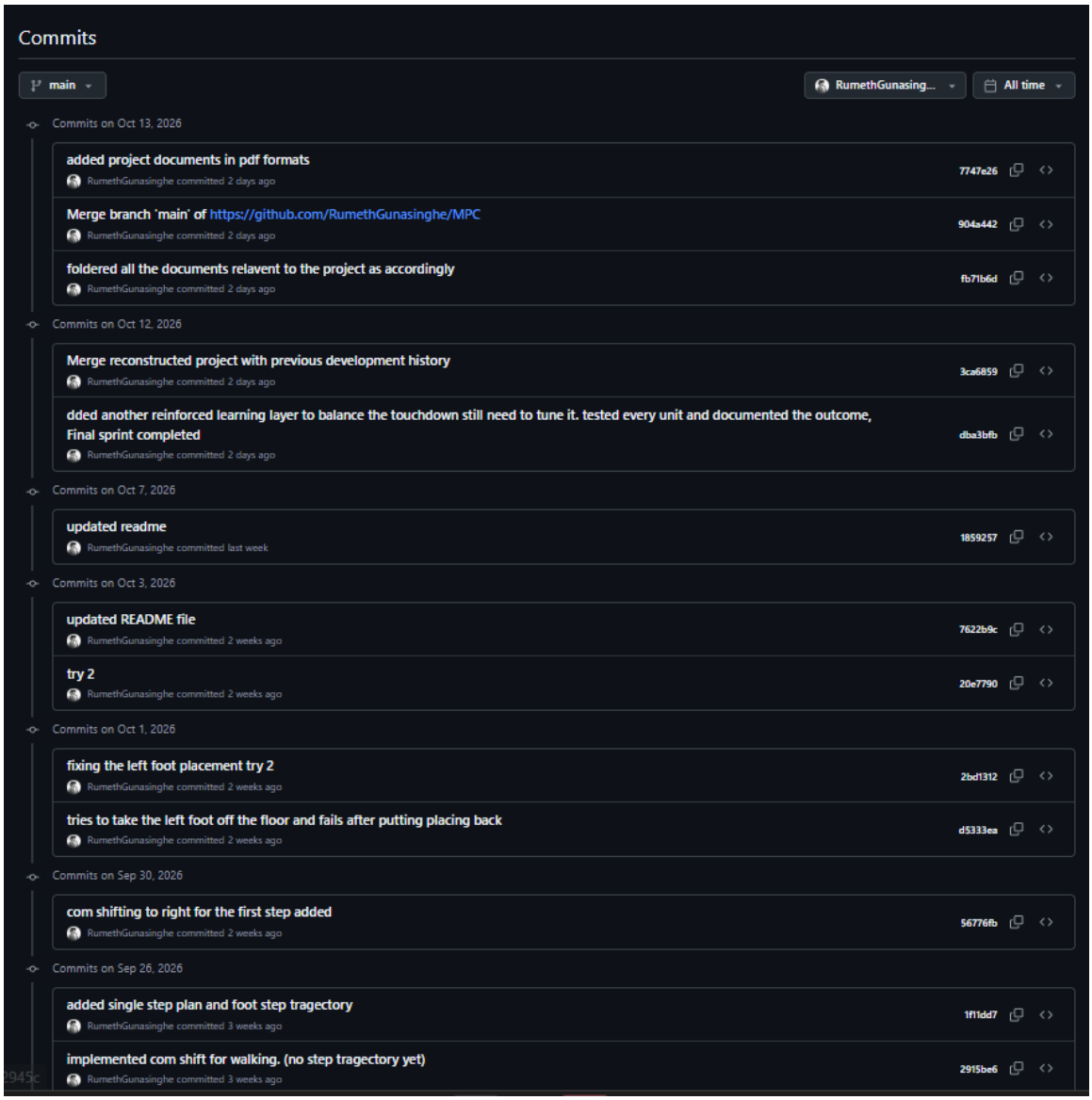
- Relevant client emails.



- Supervisor meeting/email correspondence.



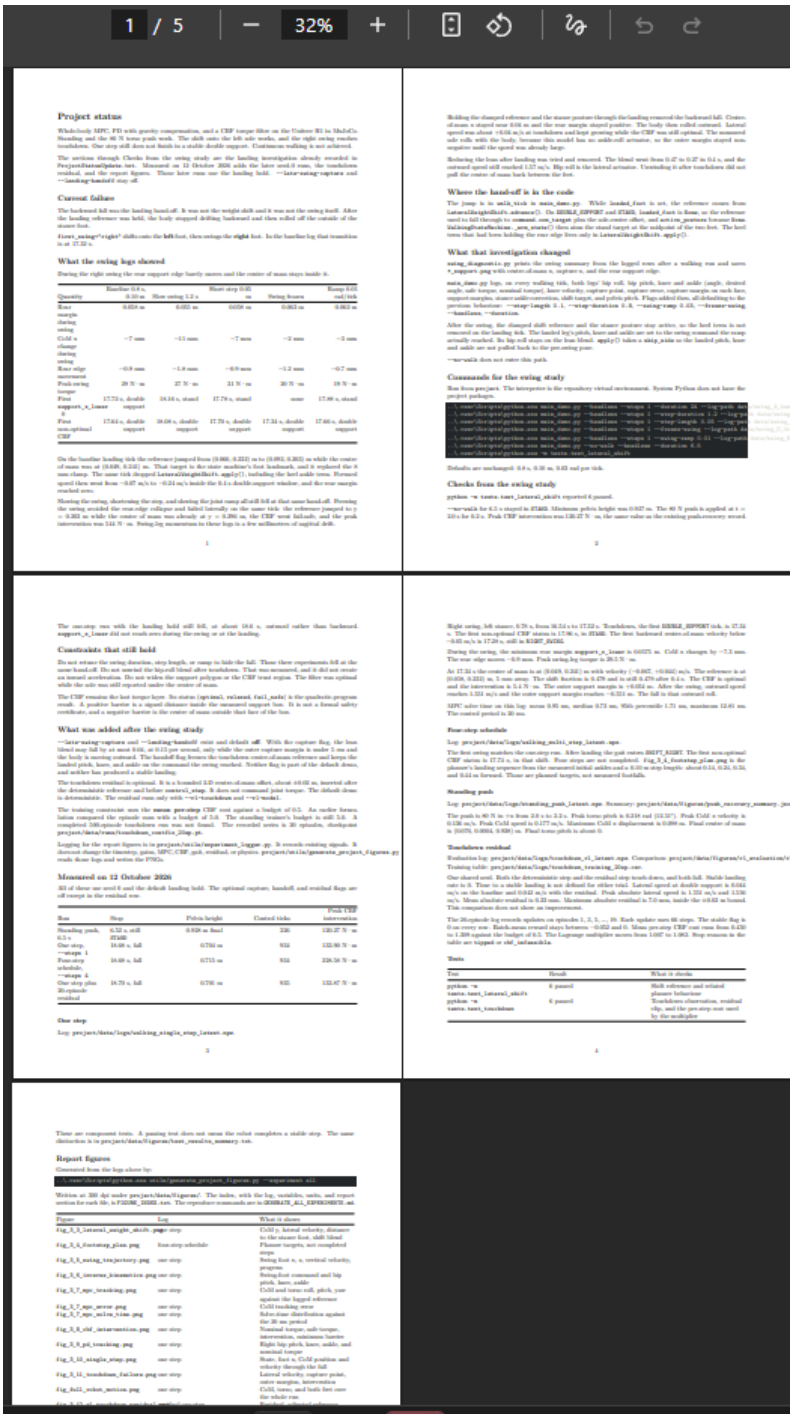
- GitHub collaboration history.



- Team meeting notes.

Meeting	Date	Time	Attendees	Detailed Agenda & Discussion	Demonstration / Evidence Presented	Supervisor Feedback	Improvements Required	Actions & Outcomes	Evidence To Attach
1	05-Aug-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Introduced project objectives for Whole-Body Model Predictive Control on the Unitree H1 humanoid. Discussed project scope, expected deliverables, research direction, client communication status, sprint structure and risk management. Reviewed challenges of humanoid balance, whole-body control and safety constraints.	Presented initial project plan, literature search strategy and proposed architecture.	Supervisor advised starting with simplified models before full whole-body control and maintaining weekly progress evidence.	Create project plan, establish Jira workflow, contact client, begin literature review.	Research direction approved. Sprint objectives defined.	Project plan, Jira board, emails
2	12-Aug-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Reviewed collected papers on MPC, LIPM, ZMP, Capture Point and humanoid balance control. Compared different control approaches and discussed suitability for	Demonstrated literature review progress and preliminary architecture diagram.	Supervisor recommended validating concepts in simulation before implementing advanced controllers.	Expand literature review and justify chosen MPC approach.	Decision made to use LIPM as an intermediate validation model.	Research report draft, paper summaries
3	19-Aug-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Focused on MuJoCo environment setup and Unitree H1 model investigation. Discussed joint hierarchy, actuators, sensors and simulation capabilities.	Demonstrated successful loading of Unitree H1 and scene configuration.	Supervisor requested validation of standing behaviour before walking experiments.	Investigate robot states, COM behaviour and joint functions.	Simulation environment validated.	Simulation screenshots, Git commits
4									
5	02-Sep-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Discussed left hip, right hip, knee and ankle testing. Analysed influence of each joint on pitch stability and balance recovery. Reviewed system identification strategy.	Presented joint response graphs and measured robot reactions to perturbations.	Supervisor noted good progress in understanding robot dynamics.	Develop reduced-order dynamic model for predictive control.	System identification completed.	Plots, analysis figures
6									
7	16-Sep-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Reviewed project delays caused by illness and communication difficulties. Discussed sprint progress, missed activities and recovery strategy.	Presented completed work despite schedule disruption.	Supervisor highlighted importance of attendance, communication and stakeholder updates.	Recover schedule and provide regular progress reports.	Recovery plan agreed.	Emails, medical certificates, sprint board
8	23-Sep-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Focused on transition from standing control to walking architecture. Discussed gait generation, swing-leg motion, inverse kinematics, foot trajectories and controller	Demonstrated new walking modules and simulation structure.	Supervisor advised validating each module independently before full integration.	Implement gait testing and walking controller integration.	Walking framework approved.	Architecture diagrams, code commits
9	30-Sep-2026	8:00 PM	Rumeth, Thinidu, Dr Saeed	Reviewed standing recovery improvements. Discussed safety filter, capture-point correction, COM stabilisation and disturbance rejection. Analysed why the robot still falls after large pushes and limitations of ankle-only	Demonstrated stable standing behaviour before disturbance and disturbance-recovery tests.	Supervisor stated that the standing controller forms a solid foundation and recommended progressing toward stepping recovery and dynamic walking.	Implement stepping strategy, walking MPC and foot placement control.	Sprint 5 objectives defined and report preparation approved.	Videos, screenshots, performance plots

- Project progress documentation.



3. REFLECTION, LEARNING AND CONTINUOUS IMPROVEMENT

3.1 Technical Learning

This sprint significantly improved my understanding of the difference between **foot contact** and **dynamic support** in humanoid locomotion.

Initially, successful touchdown could appear to indicate that a step had been completed. However, the normal-force analysis demonstrated that geometric contact alone does not guarantee that the new foot is supporting the robot.

The right foot could be in contact with the ground while carrying very little load, while the left foot continued carrying most of the robot's weight.

This changed my approach to debugging from looking primarily at joint positions and contact flags toward analysing:

- Contact forces.
- Centre-of-mass velocity.
- Capture-point behaviour.
- Support polygons.
- Load transfer.
- Joint command continuity.
- Recovery dynamics.

3.2 Learning from Failed Experiments

One important lesson was that a modification can successfully remove a real fault without solving the overall system problem.

The landed-ankle correction was a genuine problem. Removing it eliminated the corresponding torque and velocity discontinuity. However, the robot still fell.

Similarly, holding the recovery hip roll reduced early unloading and extended contact duration, but the right foot still failed to take the robot's weight.

The experiments therefore helped establish a more rigorous approach to controller development.

3.3 Testing and Experimental Methodology

Another important lesson was the value of changing one variable at a time. Instead of simultaneously changing:

- MPC parameters,
- CBF parameters,

- swing timing,
- step length,
- hip-roll control,
- ankle control,

I isolated individual hypotheses. This made it possible to determine that:

- the ankle discontinuity was real,
- the hip-roll transition contributed to early unloading,
- neither modification alone produced stable double support.

This approach makes subsequent experiments easier to interpret and provides more defensible evidence for the project report.

3.4 Limitations Identified

The current walking implementation remains limited because the robot does not yet reliably transfer load to the newly landed foot.

The current recovery controller detects geometric contact, but contact detection alone does not guarantee that the landed foot becomes a load-bearing support. The current results therefore do not justify claiming successful multi-step walking.

The most important remaining technical areas are:

- Controlled load transfer during double support.
- Maintaining the landed foot as a planted contact.
- Coordinating CoM motion with support-foot loading.
- Managing lateral velocity before and after touchdown.
- Developing an appropriate double-support recovery objective.

These limitations will guide the next stage of development.

3.5 Continuous Improvement Plan

The next experiment will investigate whether gradually transitioning the recovery hip-roll command is preferable to applying the recovery capture correction in a single control tick.

The planned experiment will use a configurable hip-roll slew while keeping:

- landed-ankle capture suppression,
- MPC,
- CBF,

- support logic,
- swing trajectory,
- step length,
- and recovery-state logic

unchanged. The main evaluation criterion will not simply be whether the robot falls. Instead, the experiment will evaluate whether the new foot actually begins to carry load by examining:

  - right-foot normal force,
  - left-foot normal force,
  - right-foot contact duration,
  - time until right-foot force falls below 10 N,
  - centre-of-mass lateral velocity,
  - pelvis roll,
  - CBF intervention,
  - and stable-double-support duration.

If the hip-roll slew improves contact but the right normal force still approaches zero, the next investigation will focus on explicit planted-foot and load-transfer control.